Numerical skills for neuroscience

Part 1 · Functions, change and rhythms

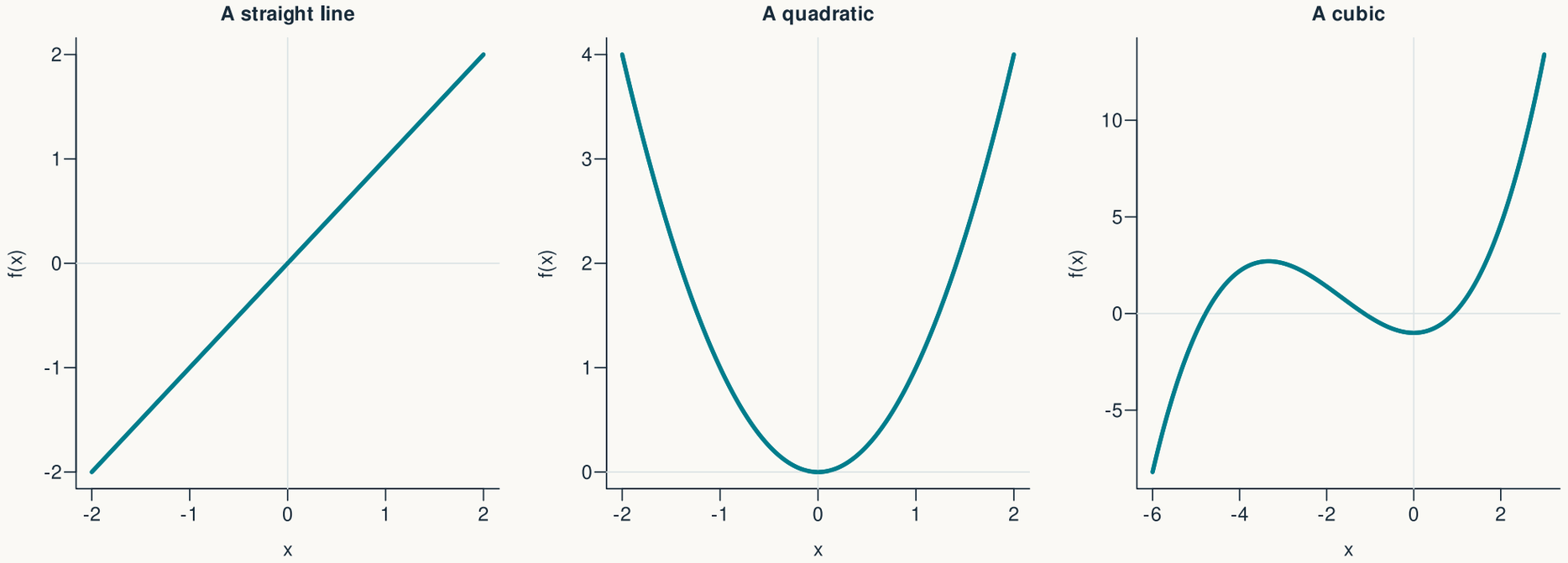
Matteo Lisi

Outline part 1

A mathematical tool	A question it helps us ask
Function $y = f(x)$	How are two quantities related?
Derivative dy/dt	How quickly does something change?
Fourier transform	Which rhythms are present?

A function maps an input to an output

$$x \longrightarrow f(x) \longrightarrow y$$



$$f(x) = x \qquad f(x) = x^2 \qquad f(x) = \tfrac{1}{5}x^3 + x^2 - 1$$

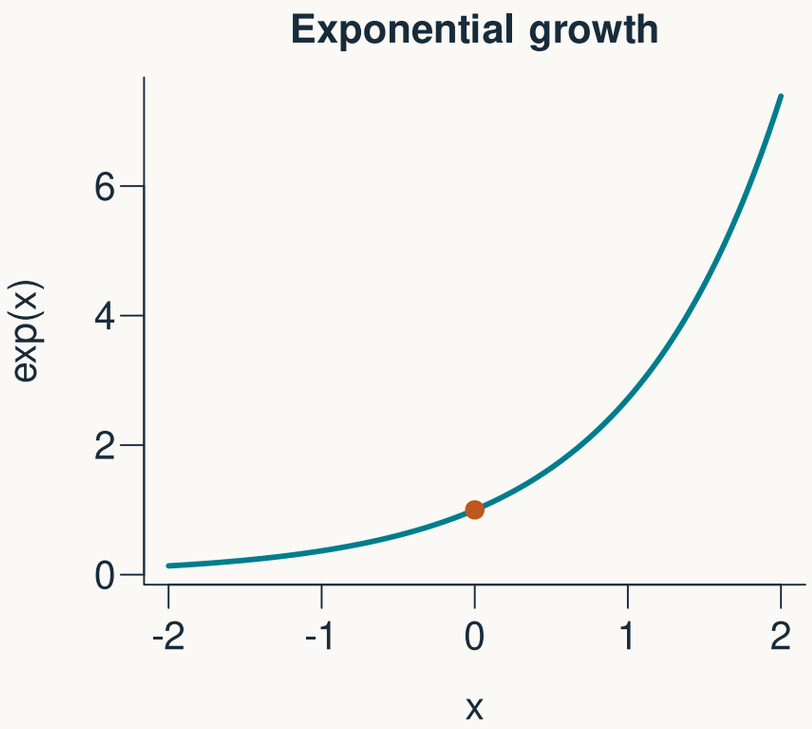
Function notation (example: exponential decay)

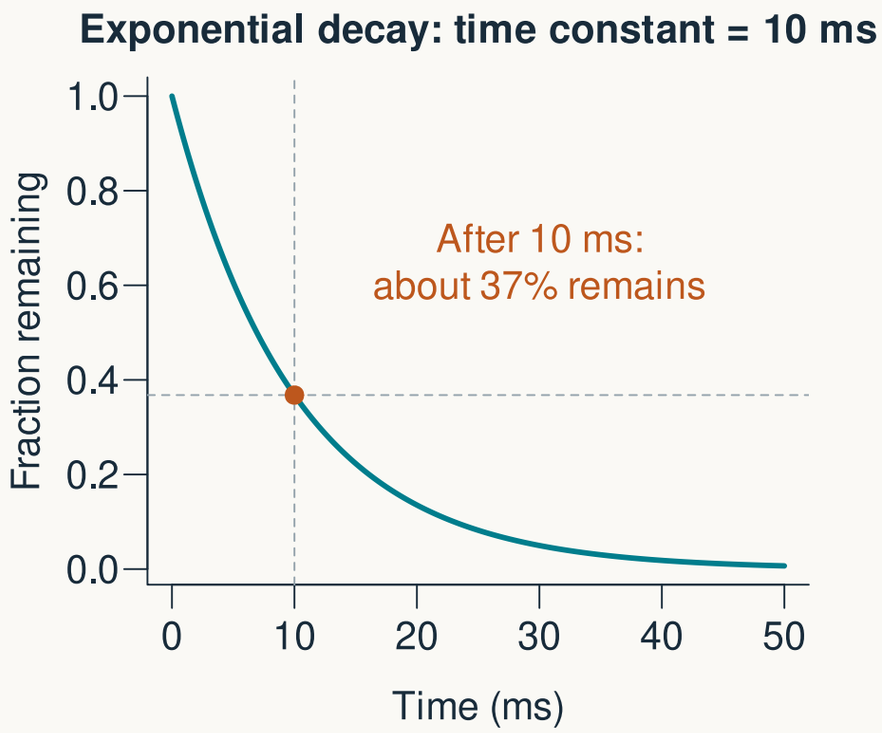
$$\underbrace{V(t)}_{\text{voltage at time } t} = \underbrace{V_0}_{\text{starting value}} e^{-t/\tau}$$

Symbol	Read it as
t	Input: elapsed time
$V(t)$	Output: voltage relative to baseline
$V_0,\ \tau$	Parameters: values that set the curve's shape

> Check the units: t and τ must use the same time unit.

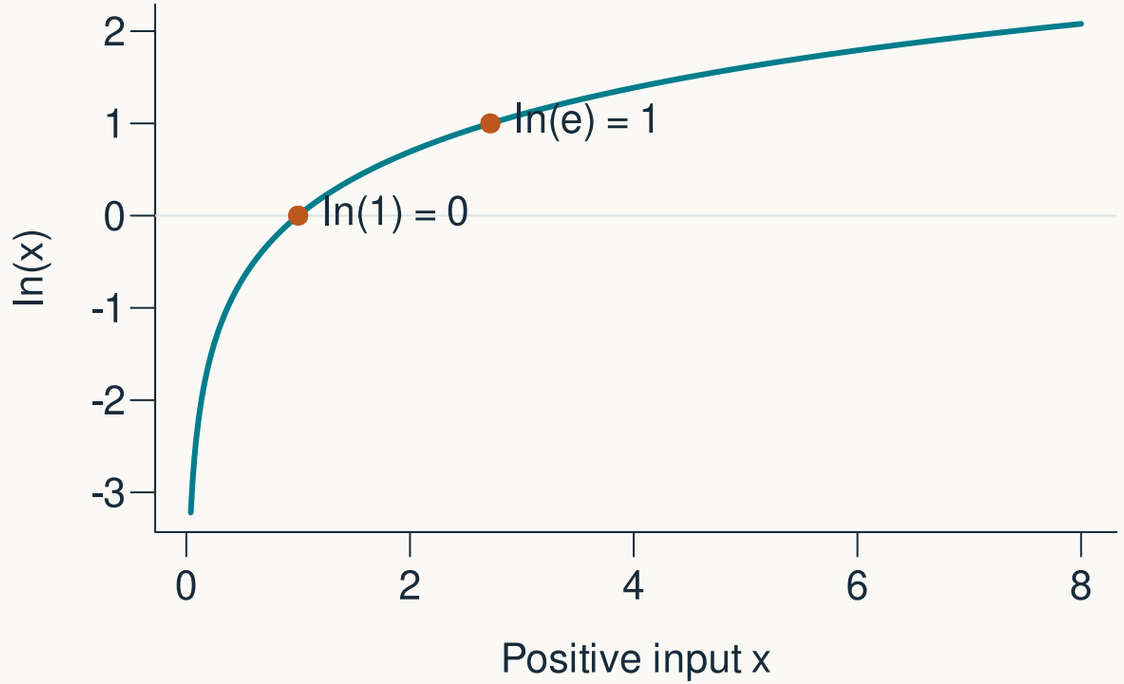
Exponentials describe growth and decay





$$\exp(x) = e^x \qquad e \approx 2.718 \qquad V(t) = V_0 e^{-t/\tau}$$

A logarithm reverses an exponential



$$\ln(e^x) = x$$

$$\ln(ab) = \ln(a) + \ln(b)$$

For real logarithms: **positive inputs**.

In R: `exp(x)` and `log(x)`.

The sum of two logarithm equal the logarithm of their product. This is useful when calculating the *joint probability* of many independent events (it is computed as the product of individual event probabilities, and it is often convenient to turn this into a sum using logarithms).

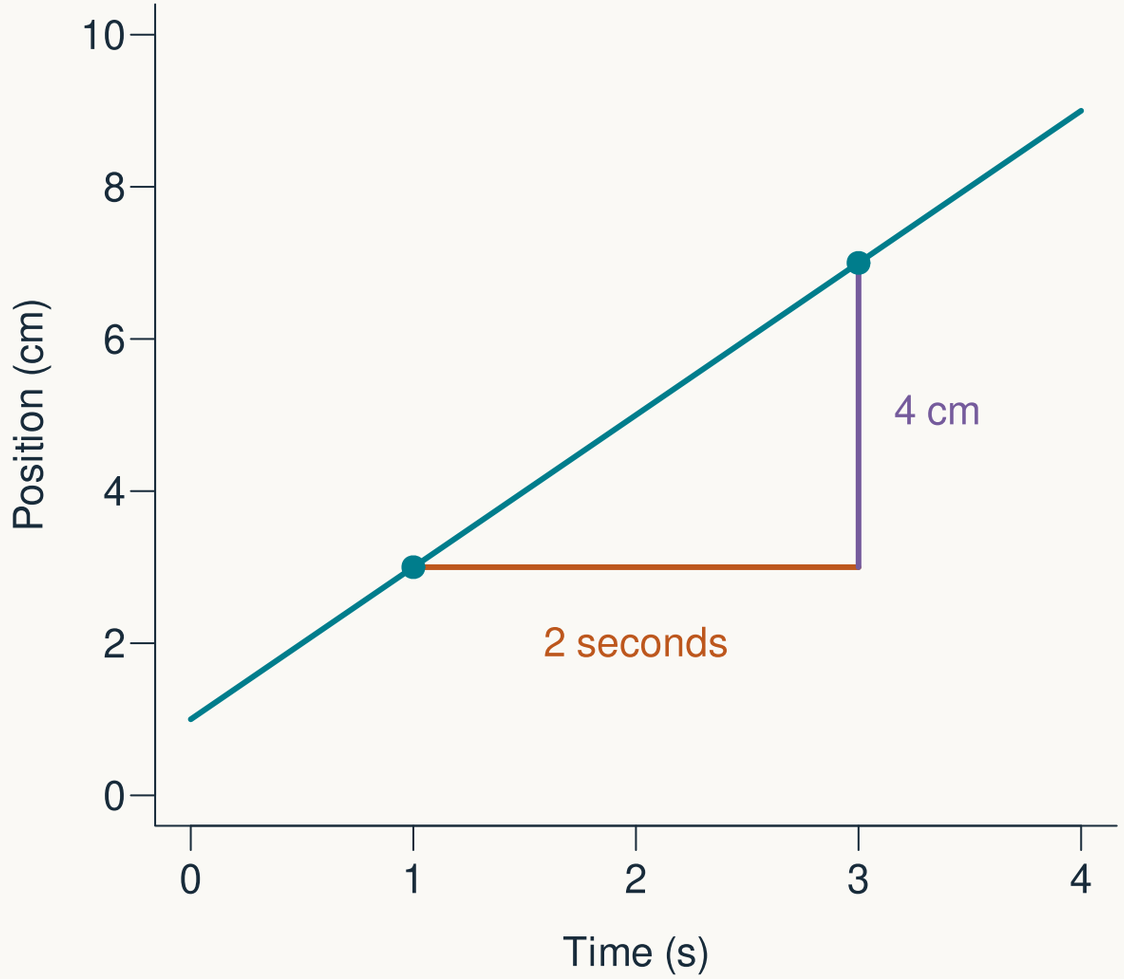
Quiz: read the curve

For $V(t) = V_0 e^{-t/\tau}$, doubling τ makes the decay…

A. faster **B.** slower **C.** unchanged

B. Slower. It takes twice as long to reach the same fraction of V_0.

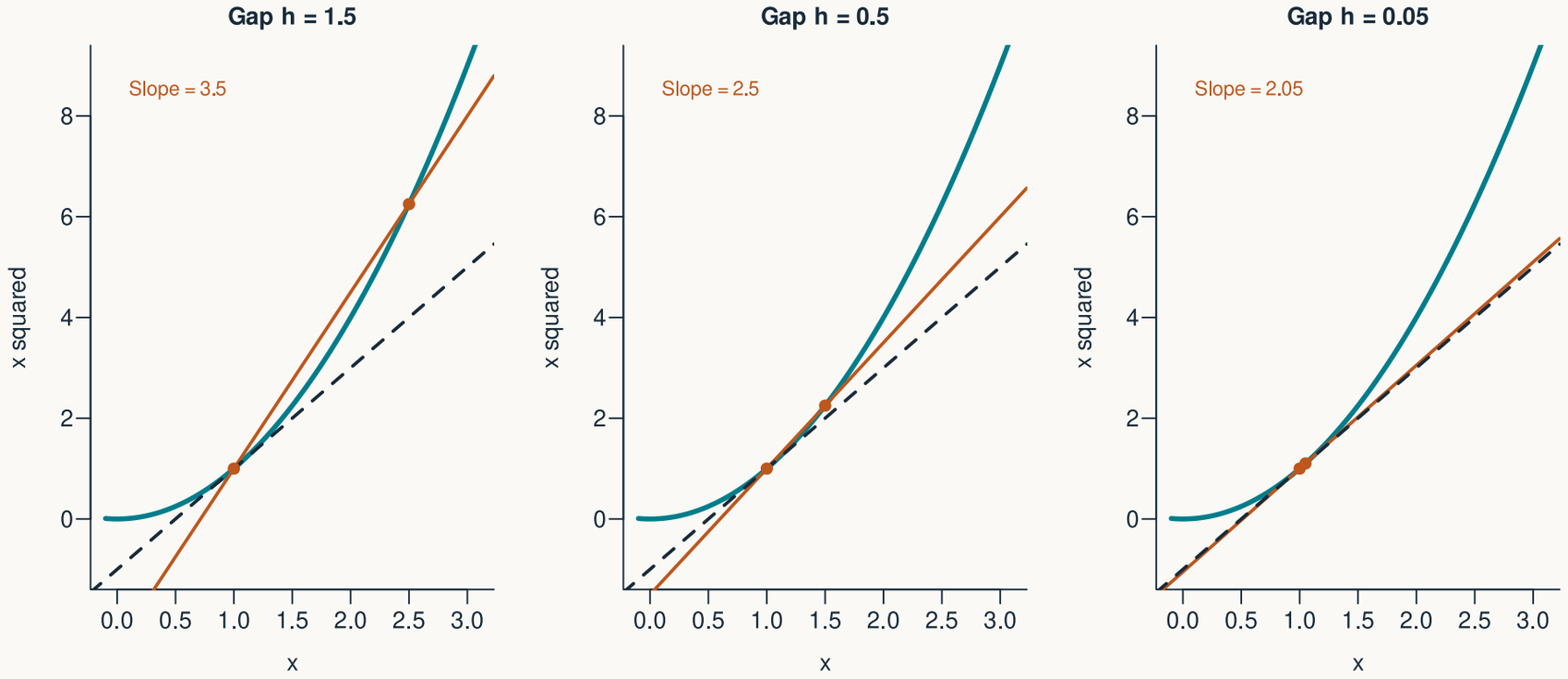
The *gradient* is a difference divided by a difference



$$\frac{\Delta y}{\Delta t} = \frac{4\ \text{cm}}{2\ \text{s}} = 2\ \text{cm/s}$$

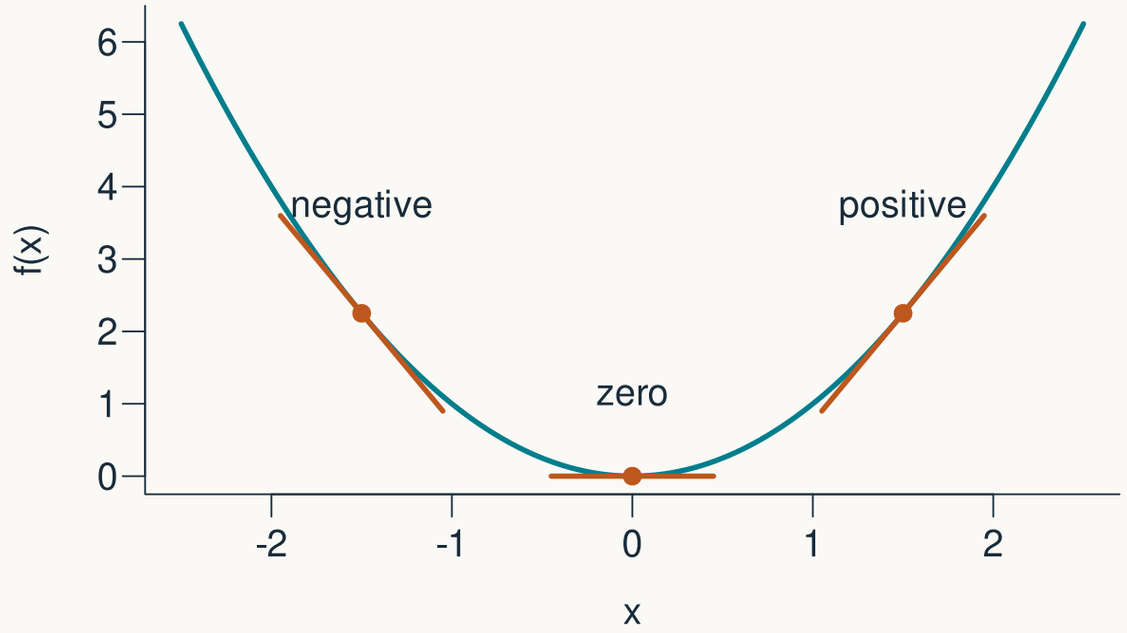
Δ means **change in**.

On a curve, the gradient depends on where you are



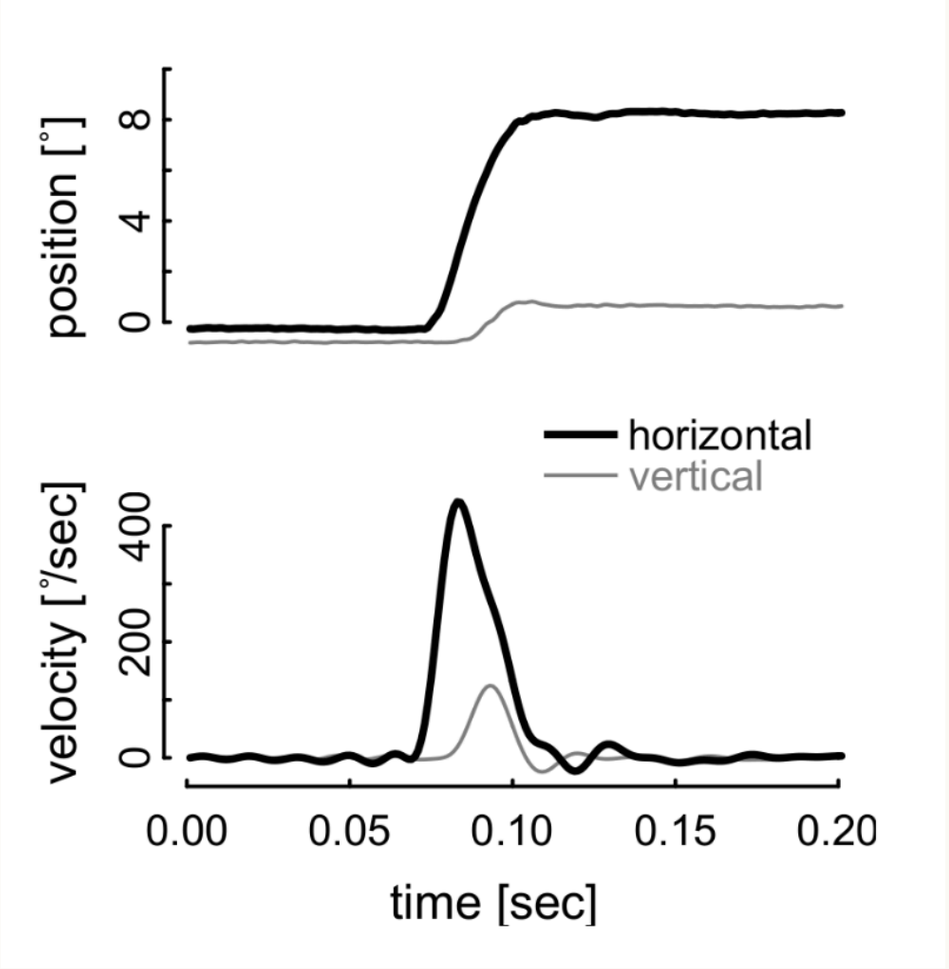
$$f'(x) = \frac{df}{dx} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

A derivative tells us the local direction of change

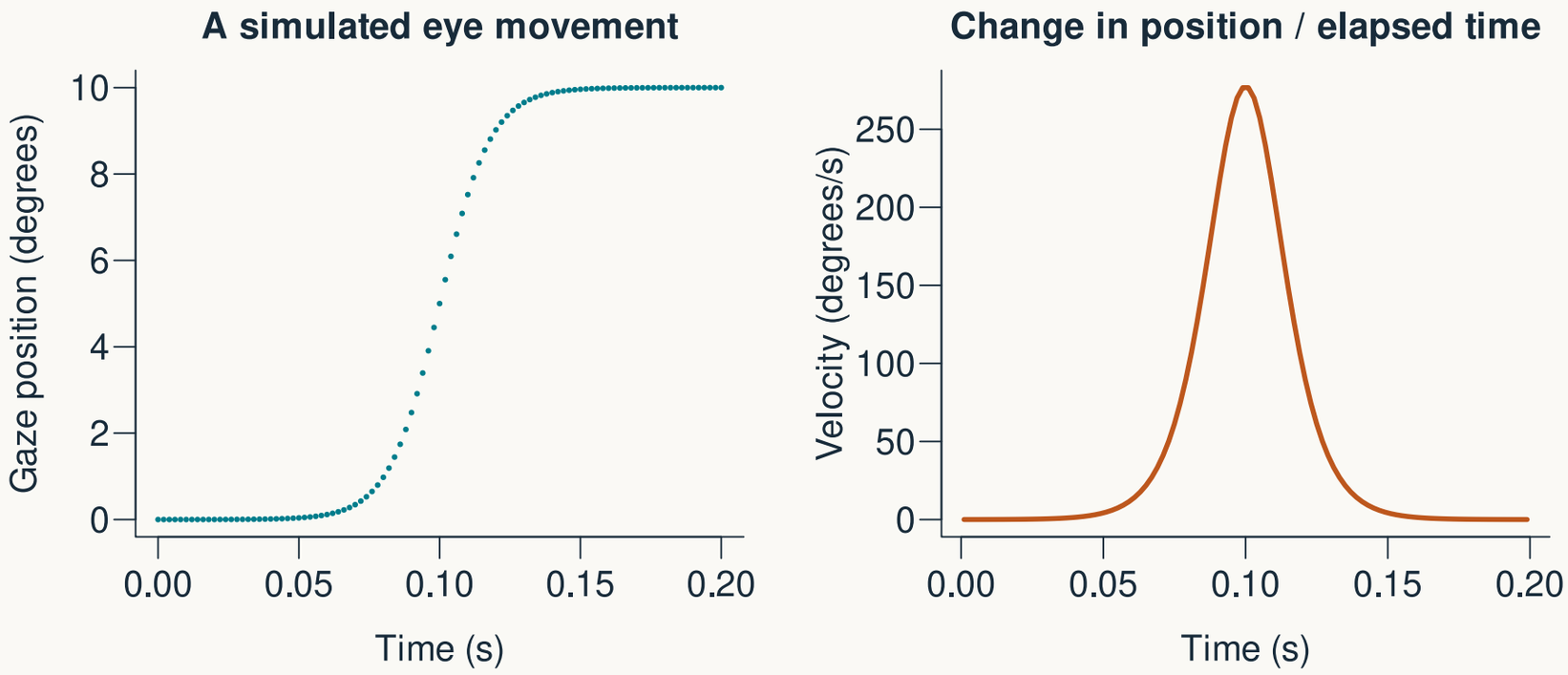


$$f(x) = x^2 \qquad \Rightarrow \qquad f'(x) = 2x$$

Example application: differentiating gaze position to compute gaze velocity



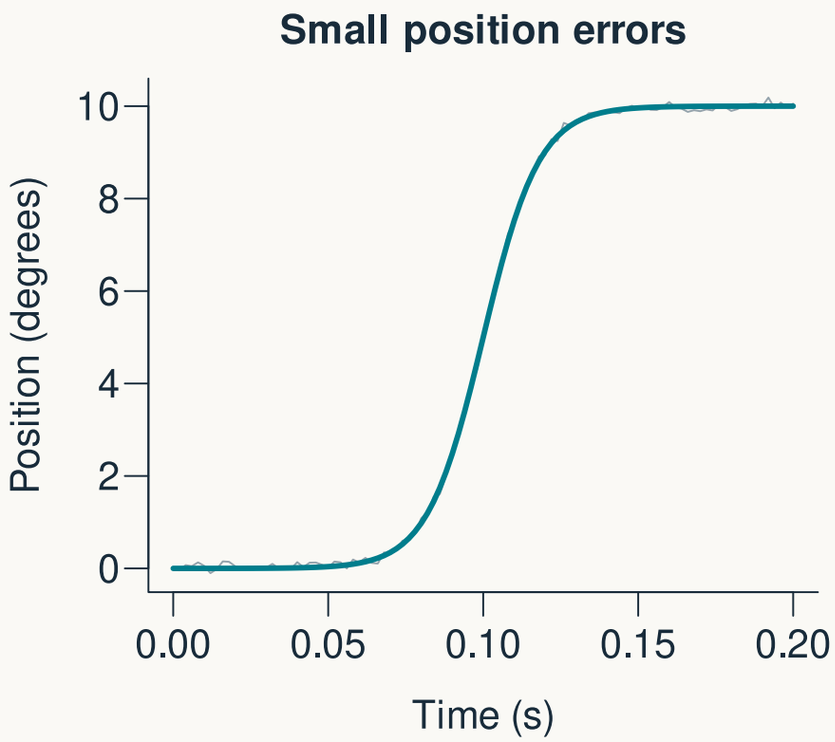
From gaze position to gaze velocity

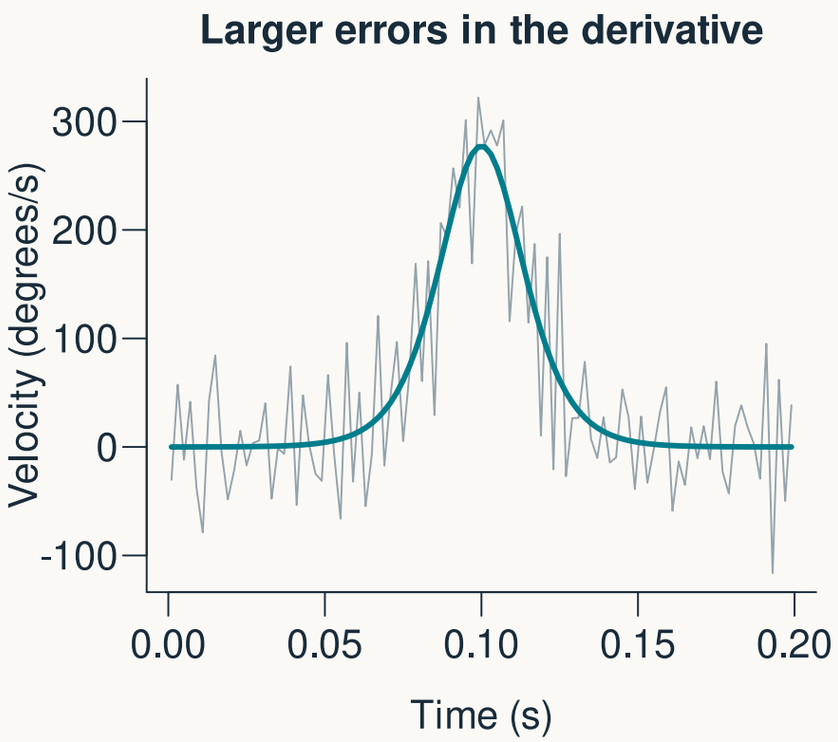


Numerical differentiation: for digital signals described by a set of points we can estimate the derivative using a finite difference approach

$$v_{i+1/2} \approx \frac{x_{i+1} - x_i}{t_{i+1} - t_i}$$

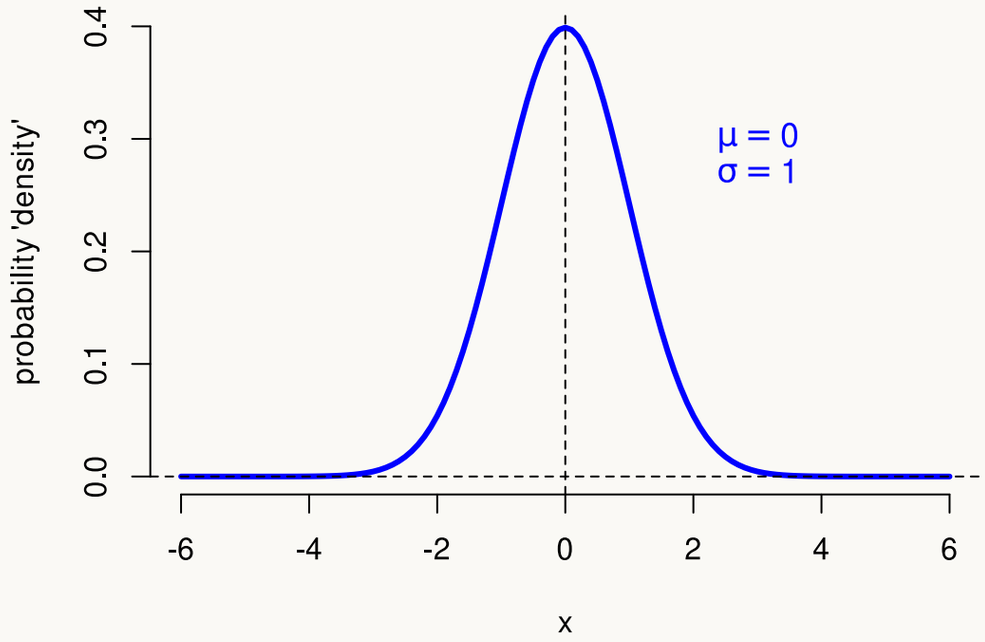
Small measurement errors can become large derivative errors





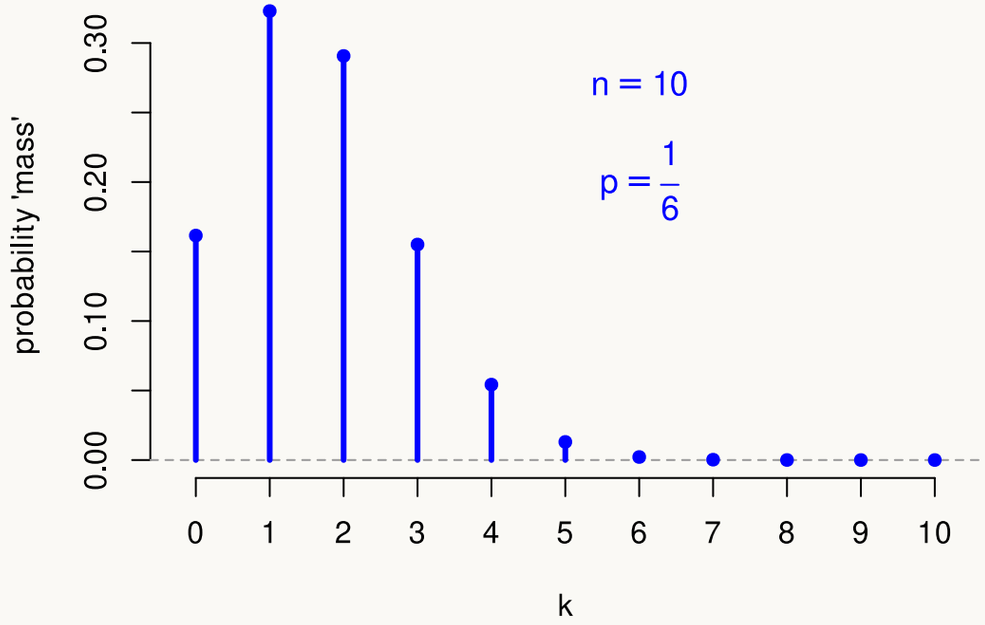
Probability *density* functions

Normal or Gaussian distribution: $p(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$



Probability *mass* functions

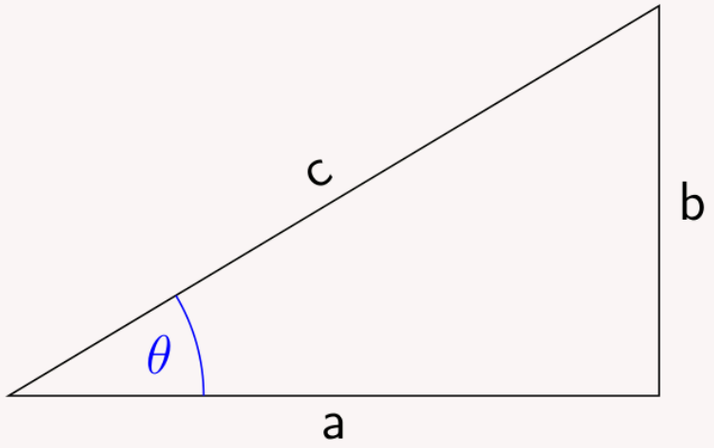
Binomial distribution: $\Pr(x) = \binom{n}{k} p^k (1-p)^{n-k}$



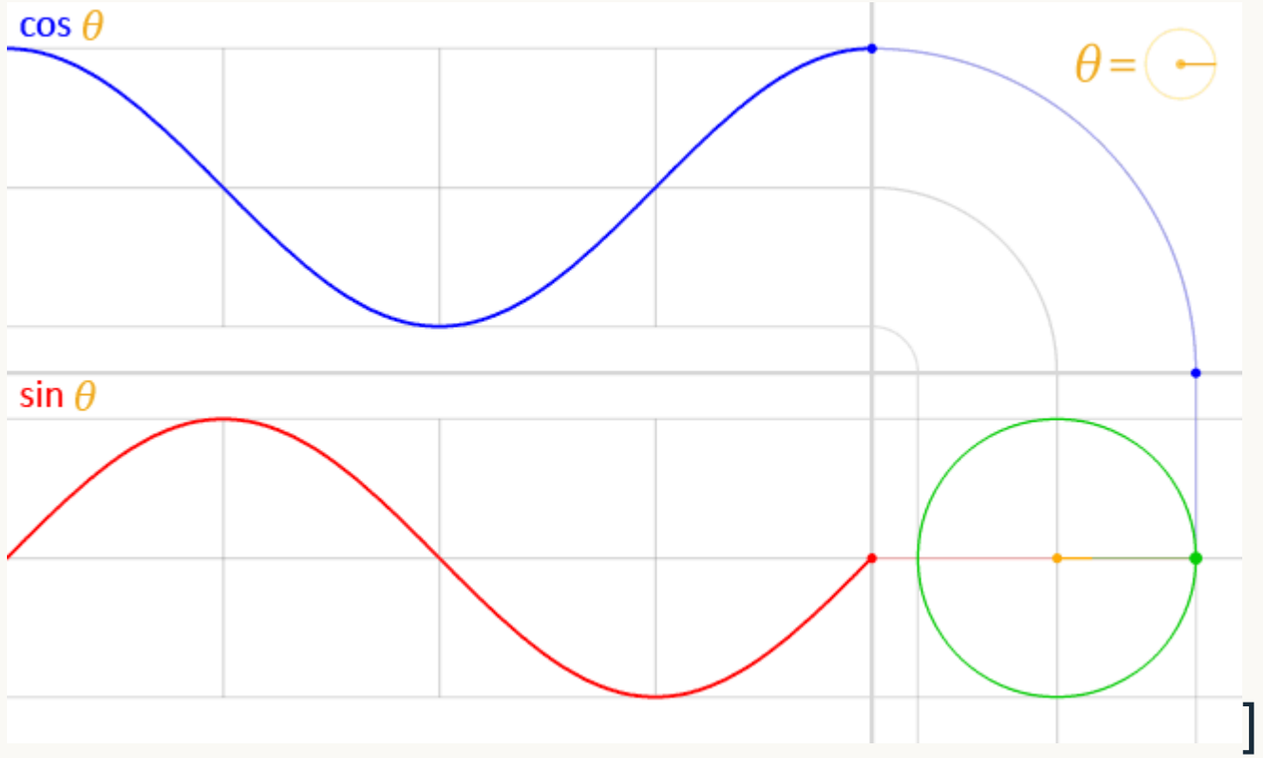
Note: this is a discrete function, and therefore not differentiable.

Trigonometric functions

Sine and cosine turn angles into oscillations

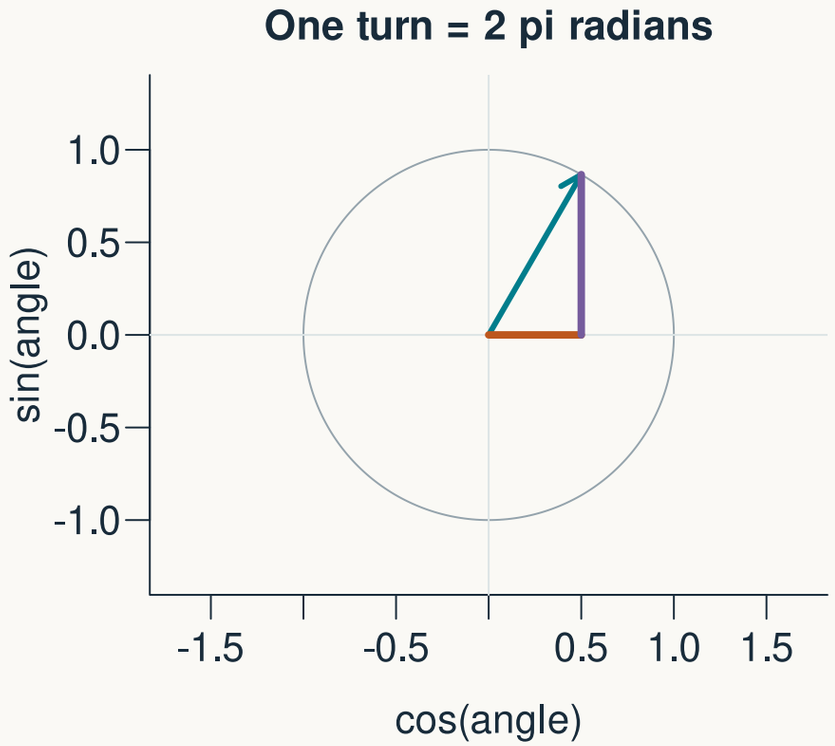


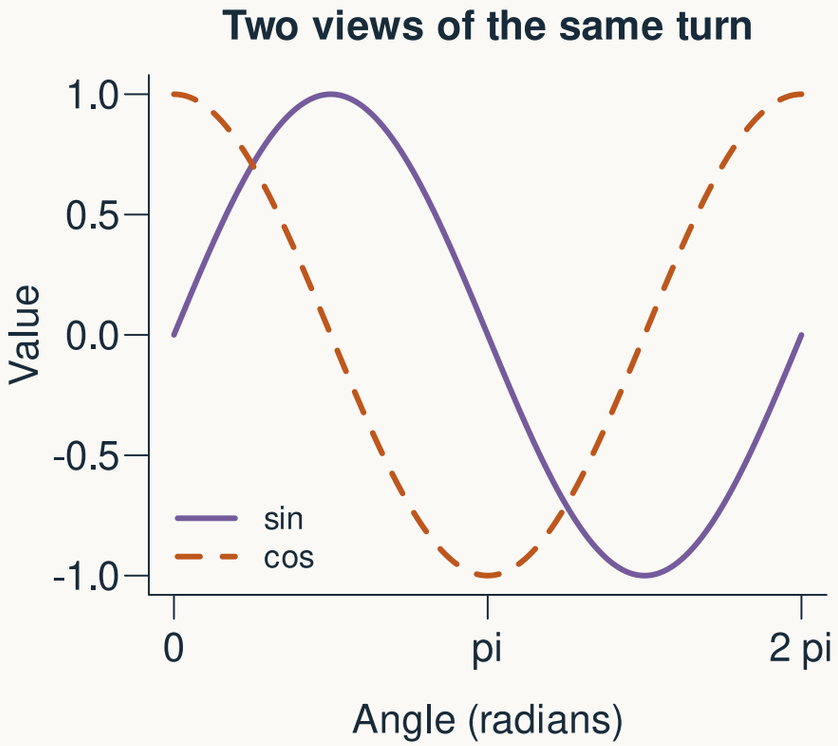
$$\sin(\theta) = \frac{b}{c} \qquad \cos(\theta) = \frac{a}{c}$$



Trigonometric functions

Sine and cosine turn angles into oscillations





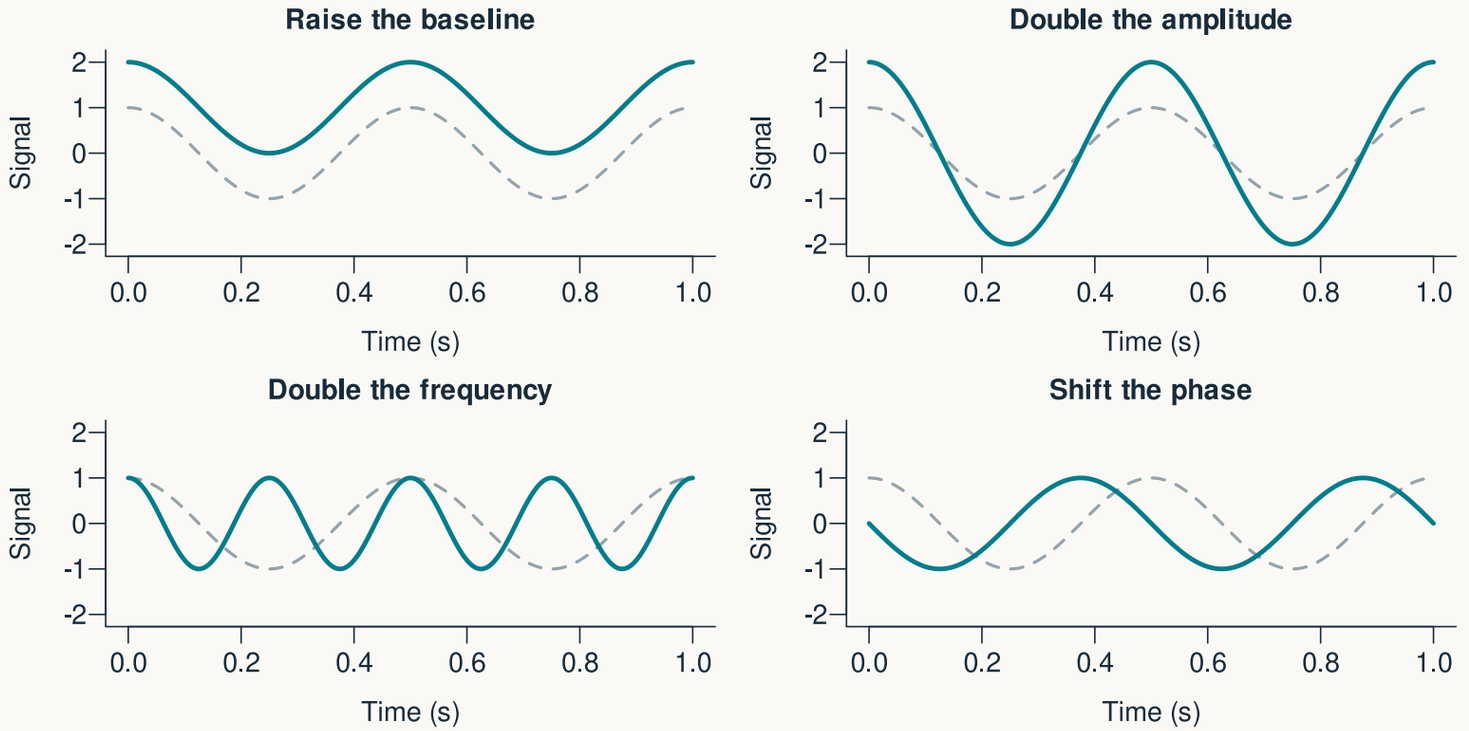
$$360^\circ = 2\pi \text{ radians}$$

4 parameters to define a rhythm

$$y(t) = b + A\cos(2\pi f t + \phi)$$

- b, average (baseline)
- A, amplitude
- f, frequency
- ϕ, phase angle

From the phase angle and frequency we can compute the time of the 1st peak: $-\frac{\phi}{2\pi f}$



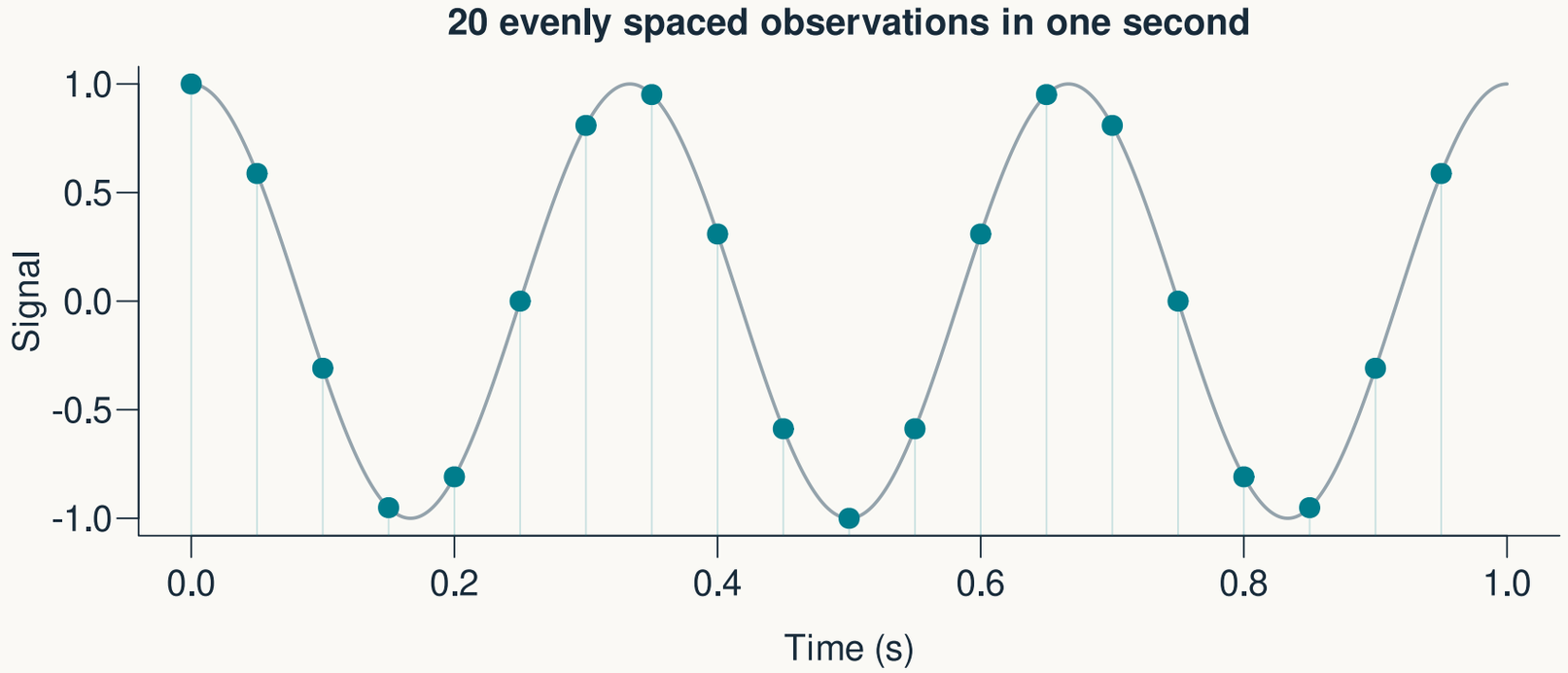
Period and frequency are reciprocals

$$f = \frac{1}{T}$$

One cycle takes…	Frequency	Units
0.1 seconds	10	Hz = cycles/second
1 day	1	cycle/day
28 days	$1/28 \approx 0.036$	cycles/day

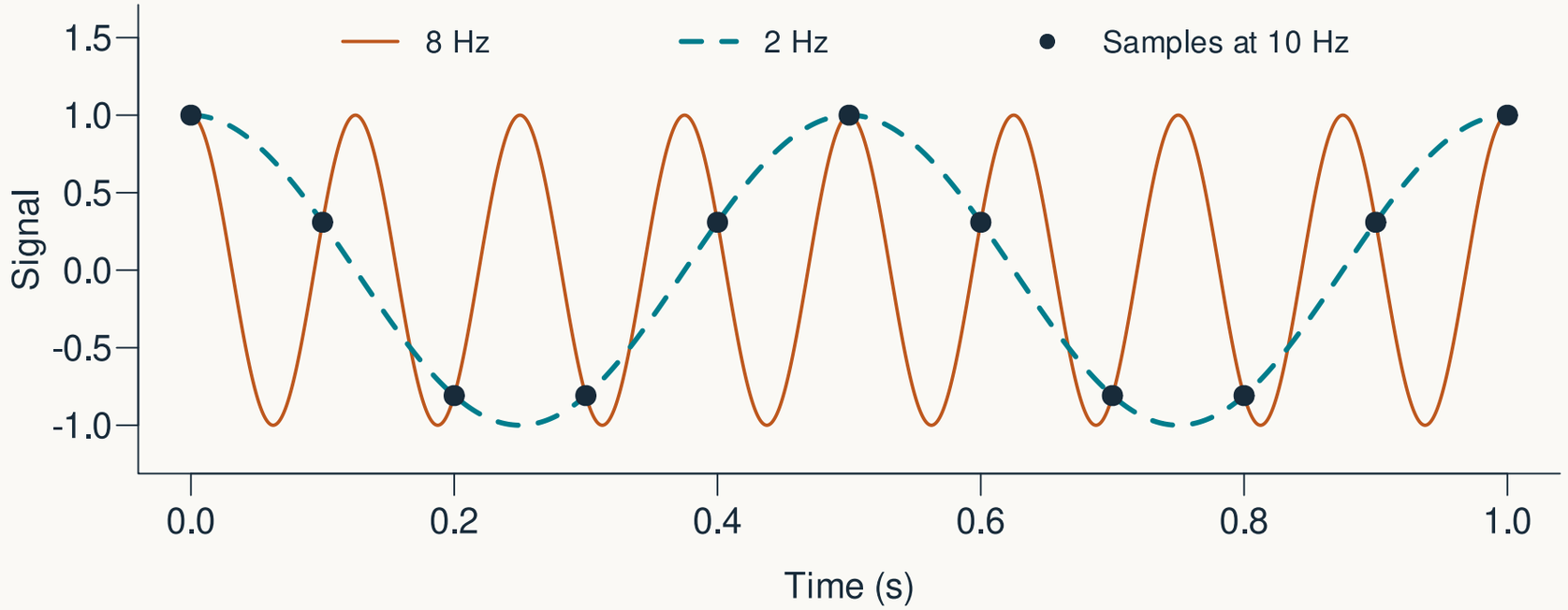
> If the period halves, what happens to the frequency?

A recording is a sequence of samples



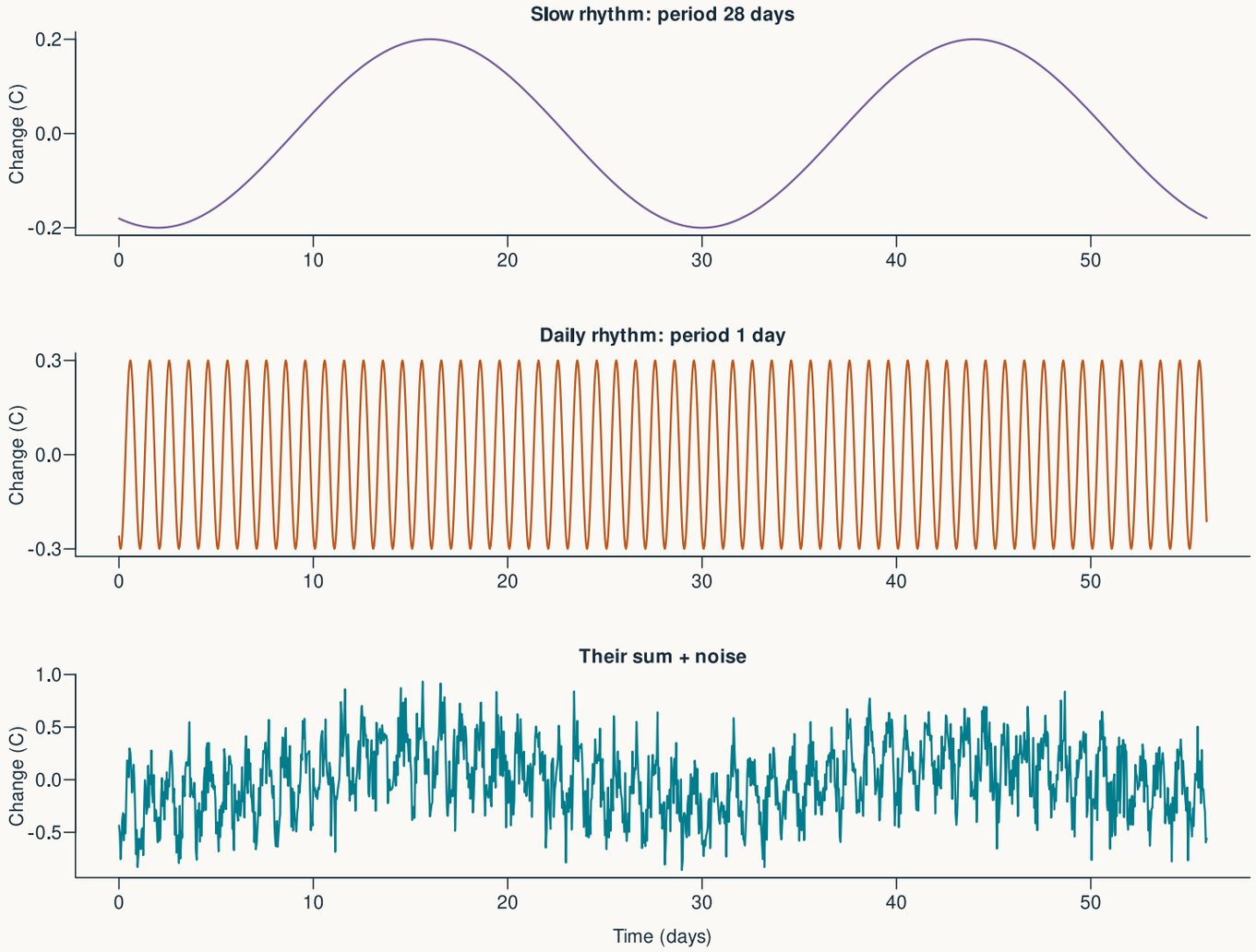
$$f_s = \frac{1}{\Delta t} \qquad t_i = \frac{i}{f_s}, \quad i = 0, \ldots, N-1$$

Sampling too slowly can disguise a rhythm



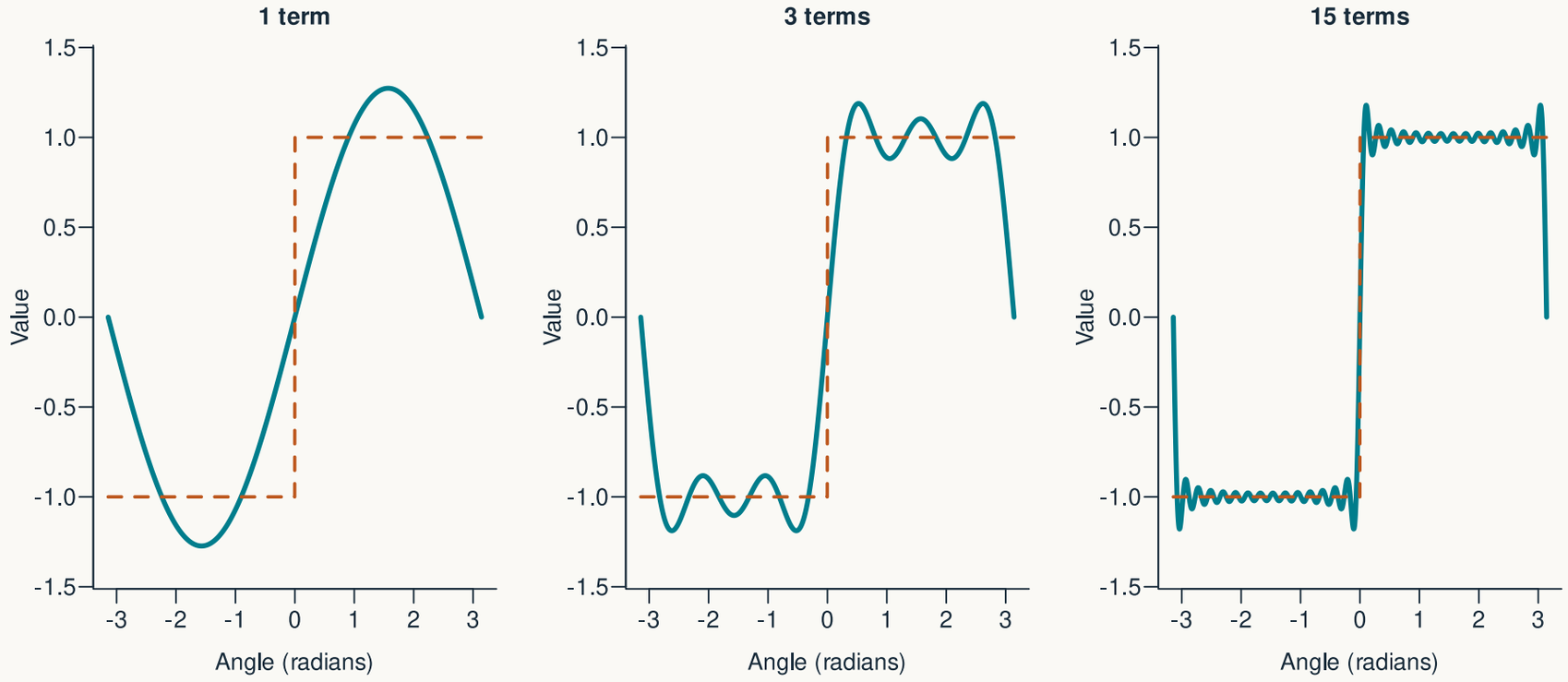
For a band-limited signal: $f_s > 2f_{\max}$, with room for anti-alias filtering.

One signal can contain several rhythms



Synthetic temperature-like signal · 56 days · hourly samples · baseline removed here

Adding sinusoids can build more complex shapes



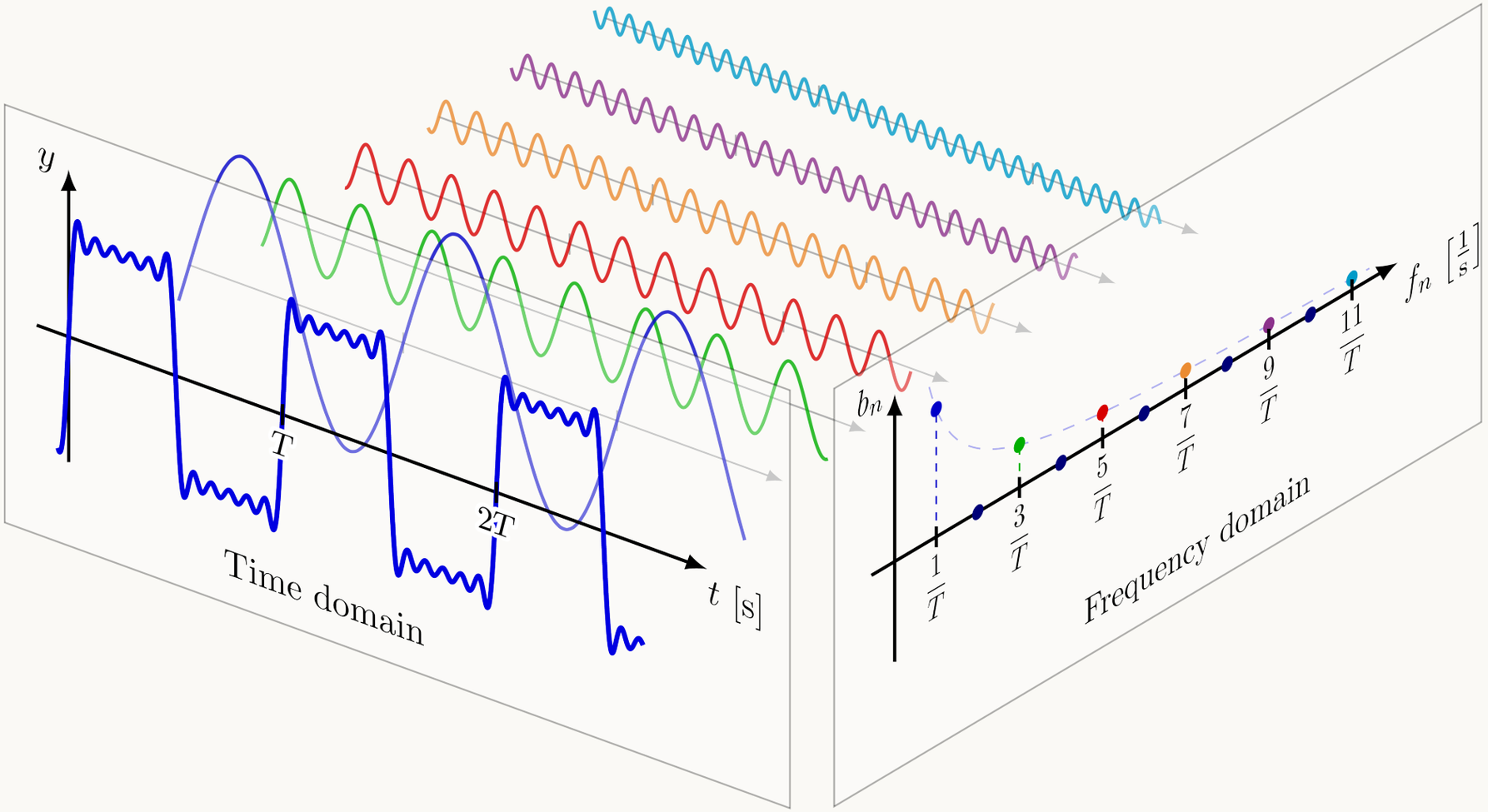
A Fourier **series** describes a periodic waveform using harmonics.

The Fourier transform changes the description

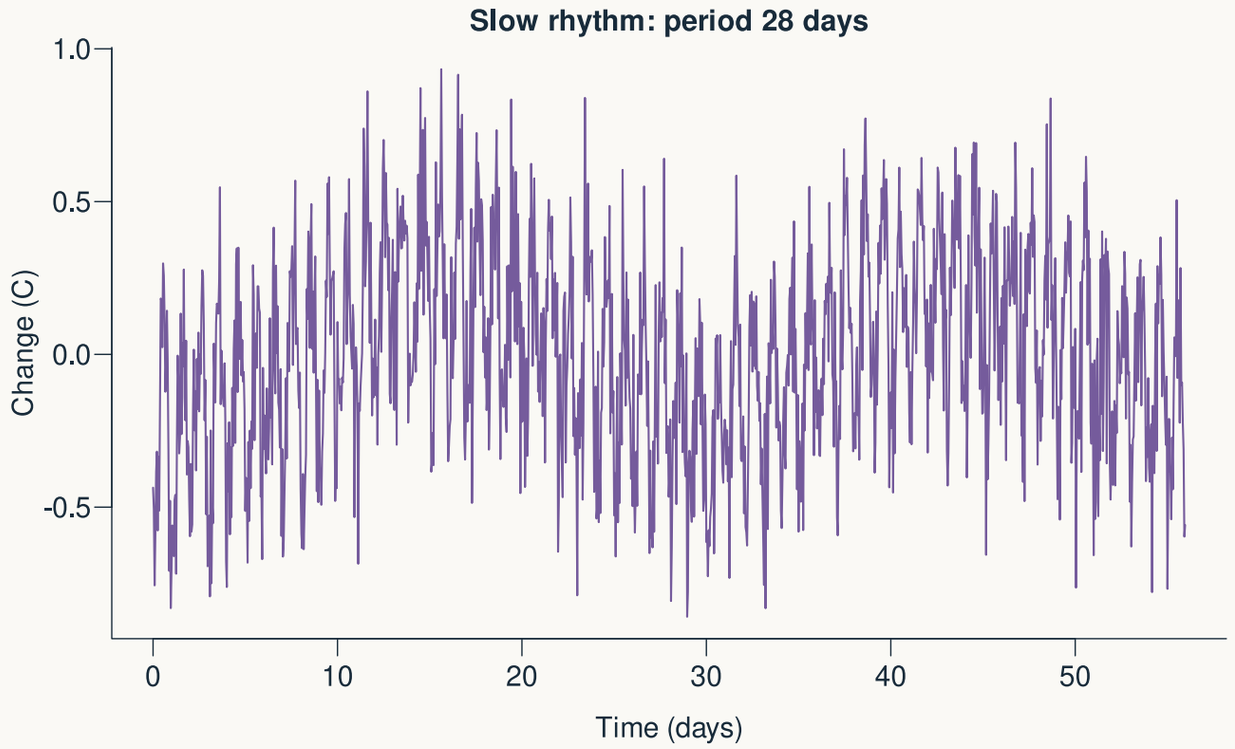
$$\underbrace{x_0, \ldots, x_{N-1}}_{\text{values over time}} \xrightarrow{\text{DFT}} \underbrace{Y_0, \ldots, Y_{N-1}}_{\text{frequency coefficients}}$$

Term in a paper	Meaning
DFT	Discrete Fourier transform of sampled data
FFT	A fast algorithm for calculating the DFT
Amplitude spectrum	Magnitudes plotted against frequency

The Fourier transform changes the description



Monthly temperature cycle



R code, Monthly temperature cycle example

```r
samples_per_day <- 24
N <- 56 * samples_per_day
t <- (0:(N - 1)) / samples_per_day

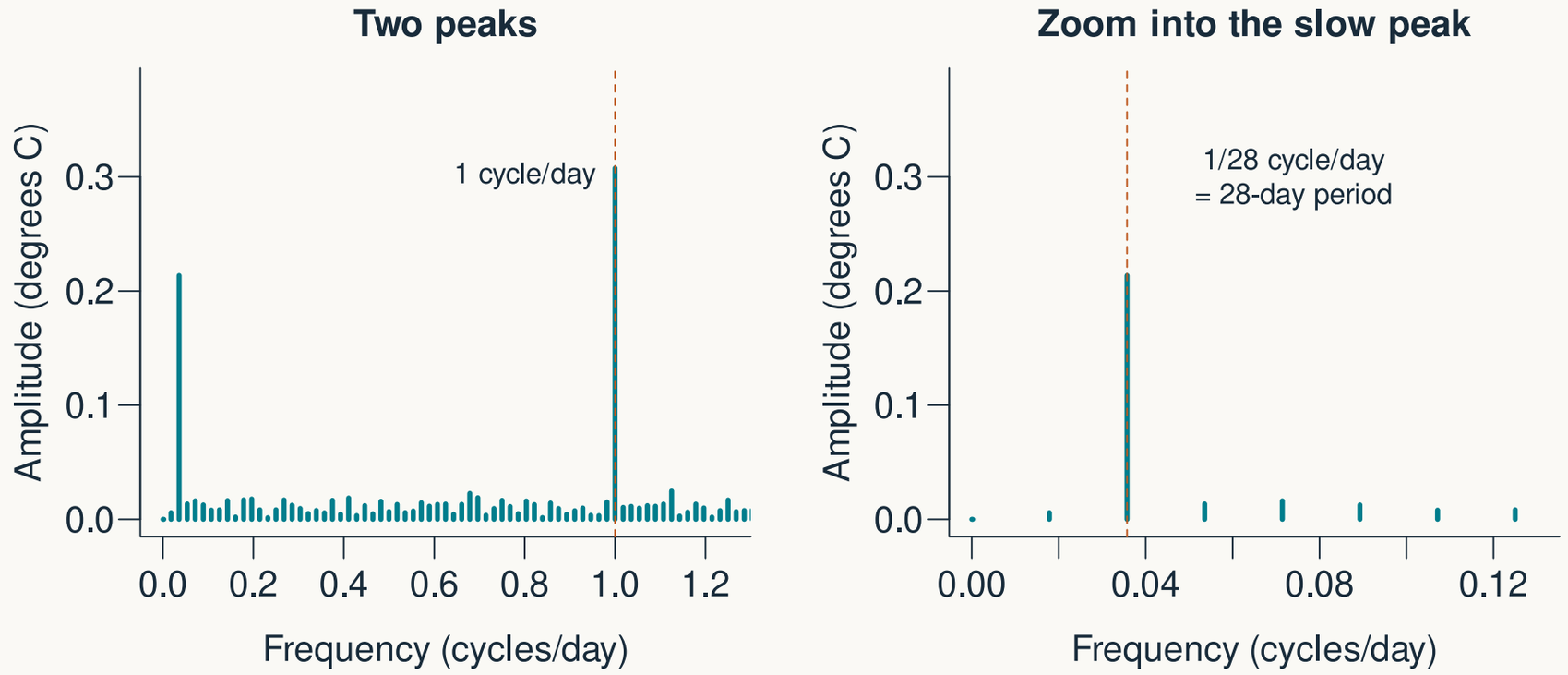
# 'pure' oscillations
slow <- 0.2 * cos(2 * pi * (t - 16) / 28)
daily <- 0.3 * cos(2 * pi * (t - 14 / 24))

# add the oscillations and introduce noise
x <- 36.8 + slow + daily + rnorm(N, sd = 0.2)

# perform FFT after subtracting mean (removing DC component)
Y <- fft(x - mean(x))
```

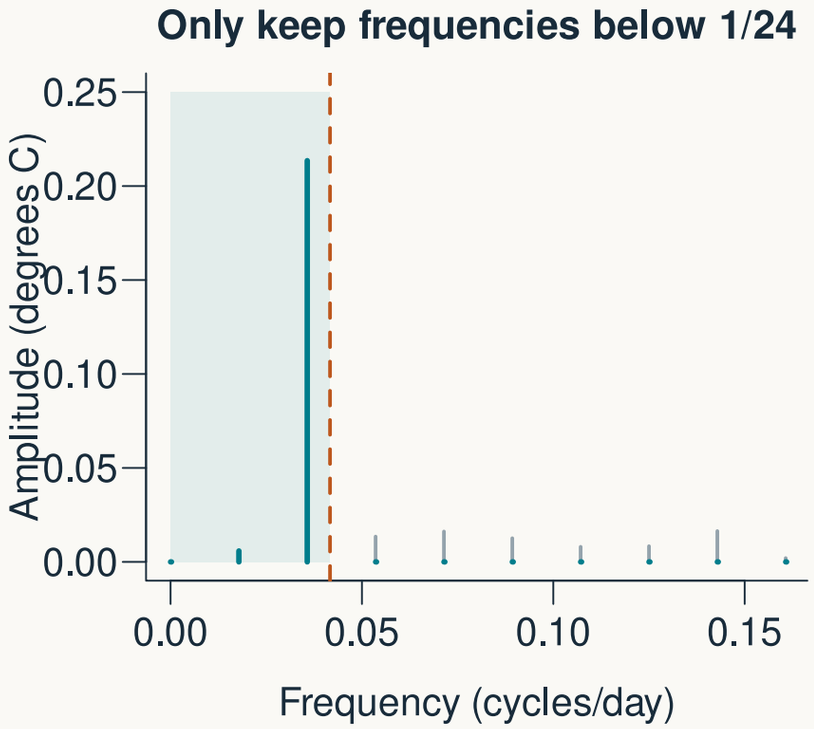
Complete, commented example: fourier_transform.R

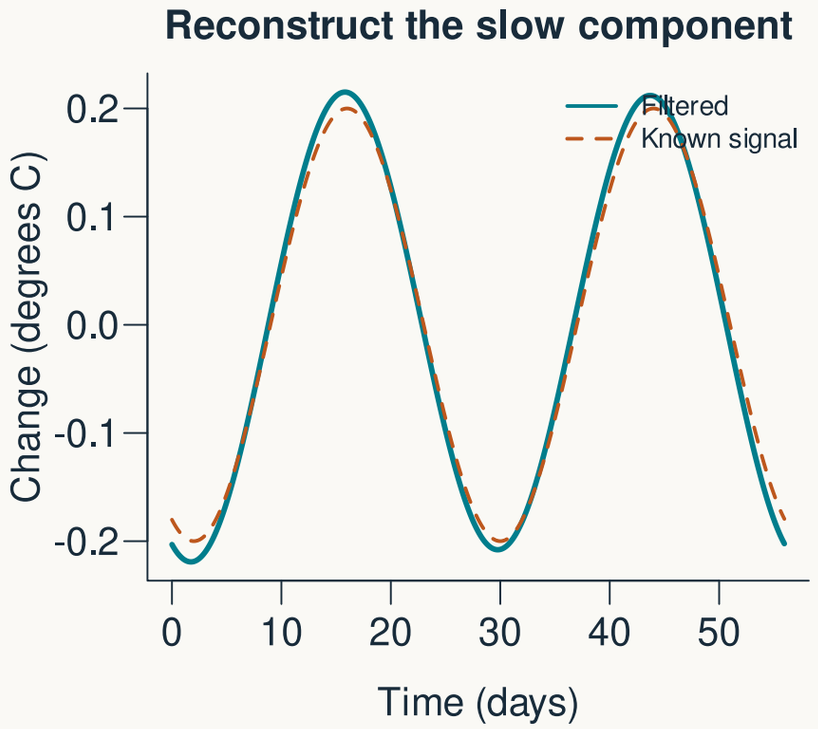
Power spectrum of monthly temperature cycle



Horizontal: **frequency**. Vertical: **amplitude**, in the signal's units.

Filtering selects which components to retain





$$|f| < 1/24 \text{ cycles/day} \iff T > 24 \text{ days} \quad (f \neq 0)$$

Translate a methods sentence

> “We sampled at 500 Hz, differentiated position, and examined the Fourier amplitude spectrum.”

500 Hz → one sample every 2 ms.

Differentiated → estimated how quickly position changed.

Amplitude spectrum → examined the strength of each frequency component.

Three things you can now explain

See this…	Ask this…
$y = f(x)$	What are the inputs, parameters and units?
dy/dt	Which change per unit time is being measured?
A spectrum	What do the axes and peaks mean?

Exit check: a peak at **5 Hz** corresponds to what period?

OPTIONAL / Images also have frequencies



Spatial frequency: **cycles per unit distance**, rather than per second.

OPTIONAL / # Differentiating the function $f(x) = x^2$

$$\frac{df}{dx} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{(x+h) - x} \tag{1}$$

$$= \lim_{h \to 0} \frac{(x+h)^2 - x^2}{h} \tag{2}$$

$$= \lim_{h \to 0} \frac{x^2 + 2xh + h^2 - x^2}{h} \tag{3}$$

$$= \lim_{h \to 0} \frac{2xh + h^2}{h} \tag{4}$$

$$= \lim_{h \to 0} 2x + h \tag{5}$$

$$= 2x \tag{6}$$

OPTIONAL /

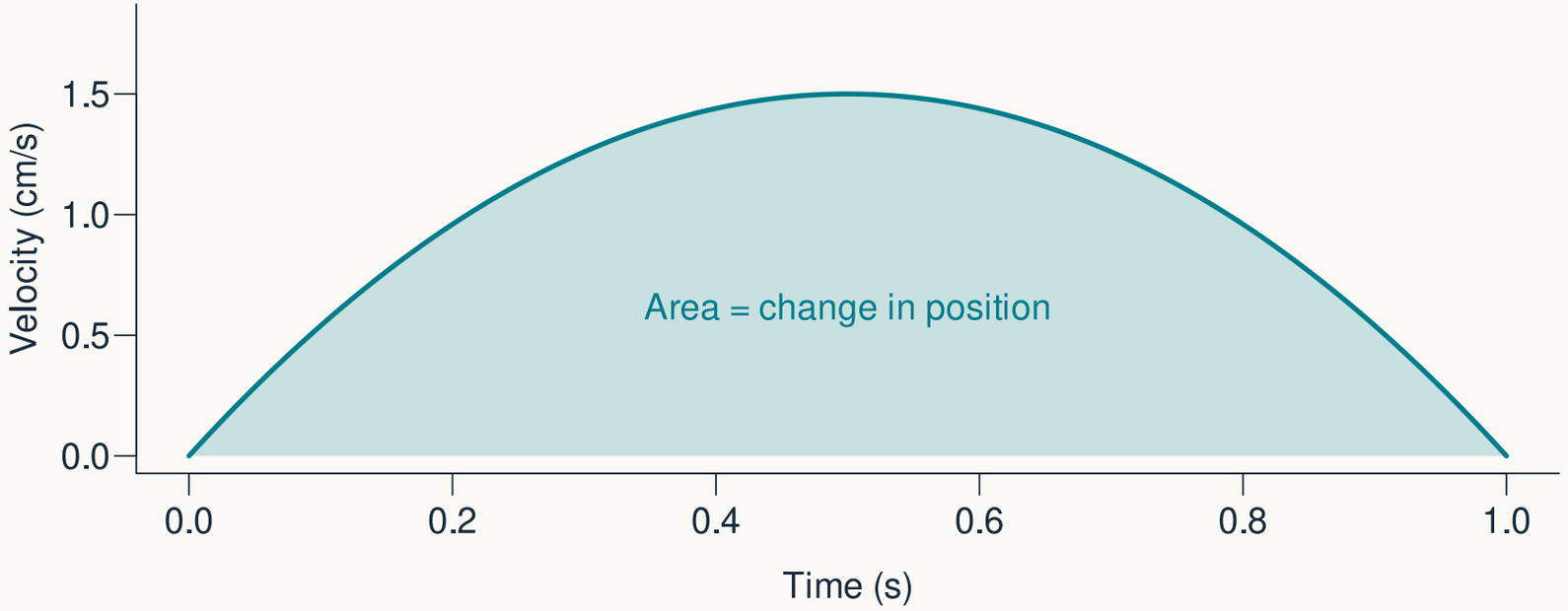
A small derivative reference

Function	Derivative
x^n	nx^{n-1}
e^x	e^x
$\ln x,\ x > 0$	$1/x$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$

$$\frac{d}{dt}e^{-t/\tau} = -\frac{1}{\tau}e^{-t/\tau}$$

OPTIONAL /

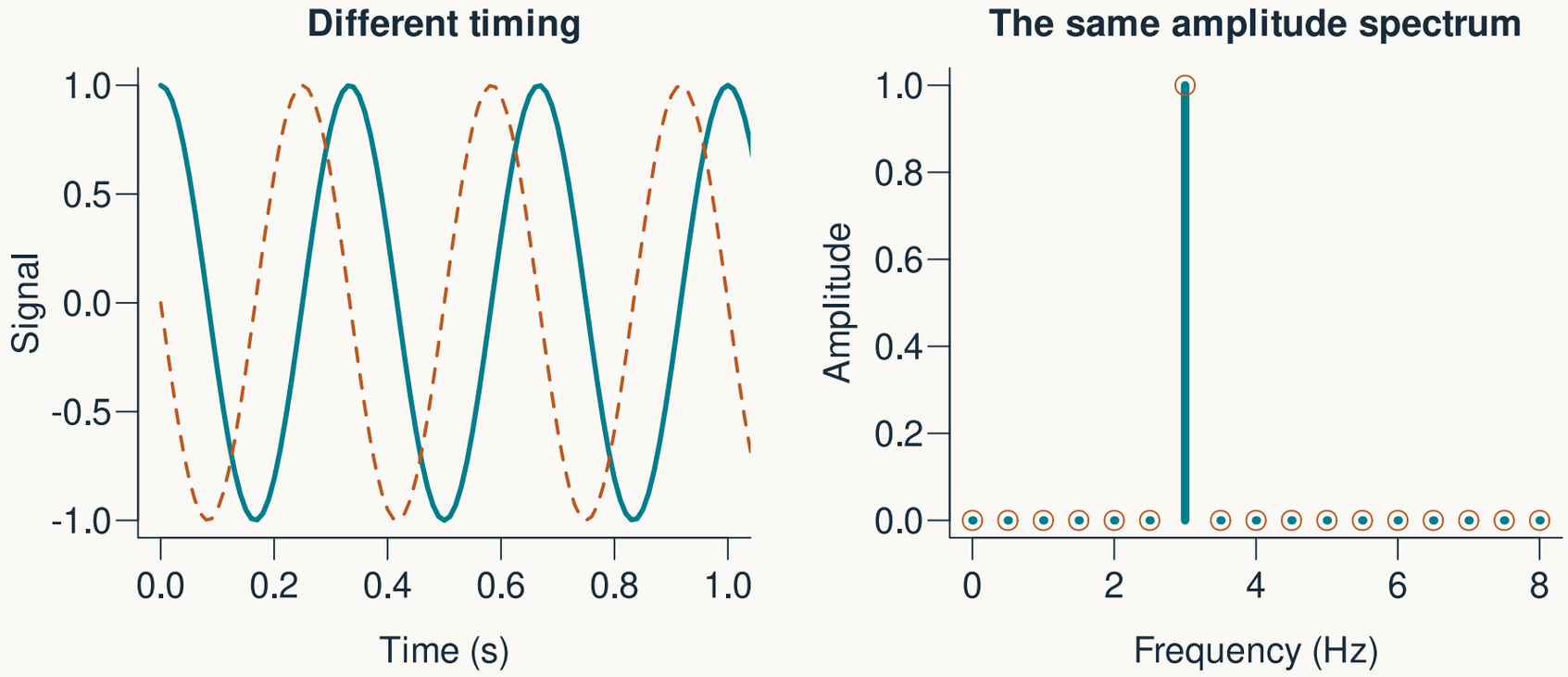
Integration adds up change



$$x(t_2) - x(t_1) = \int_{t_1}^{t_2} v(t)\,dt$$

OPTIONAL /

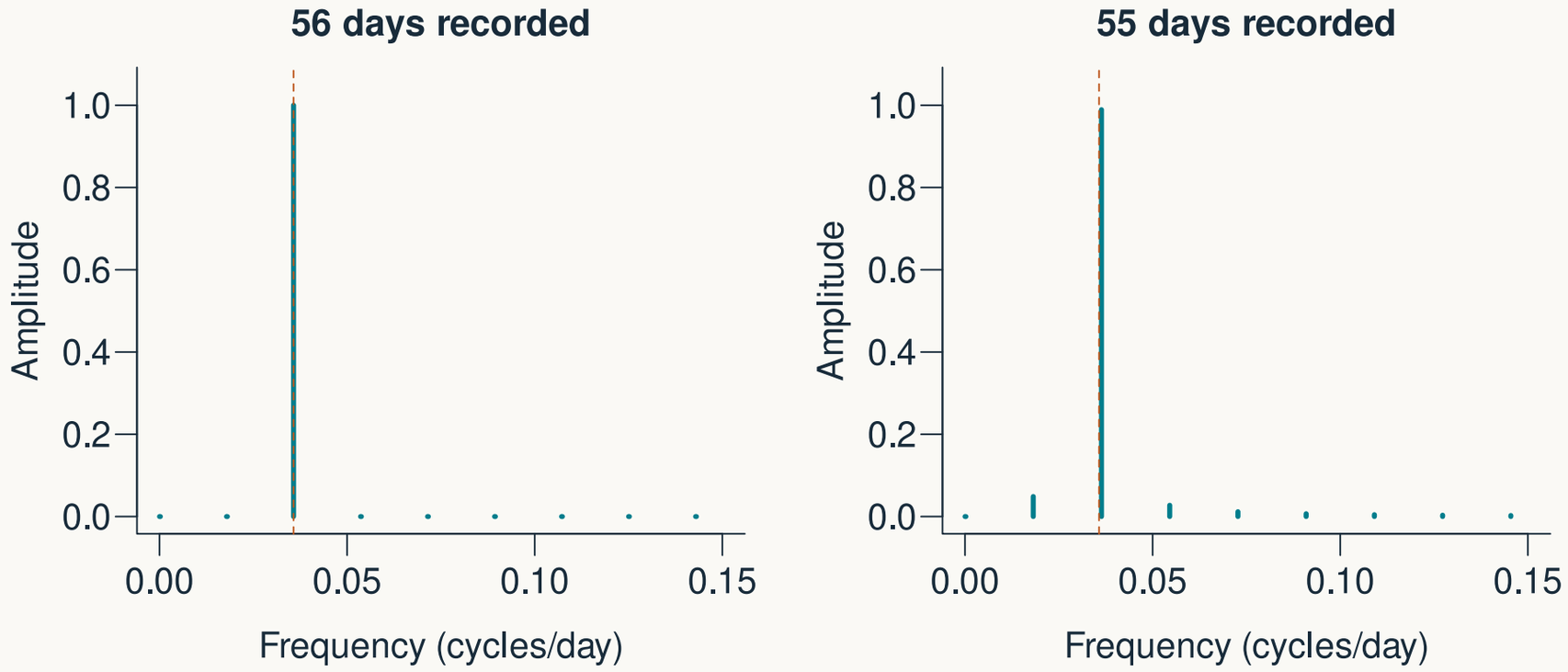
An amplitude spectrum does not show timing



> Reconstruction needs **phase** as well as amplitude.

OPTIONAL /

Finite windows spread spectral energy



> Spectral leakage depends on the recording window.